

Information Using the K 4 Buttons

Discharge Temperature (K -4 5x): _____
Target Discharge Temperature (K -4 10x): _____
Protection Mode Reading (k – 4 13x): _____
Suction Pressure(K -4 20x): _____
Suction Pipe Thermistor (K -4 19x): _____
Ambient AirThermistor (K -4 4x): _____

Suction Pipe 1 Superheat : _____
Suction Pipe Temperature _____
Minus
Low Side Saturation Temperature _____

Liquid Pipe 1 Sub-Cooling : _____
High Side Saturation Temperature _____
Minus
Liquid pipe 1 Temperature _____

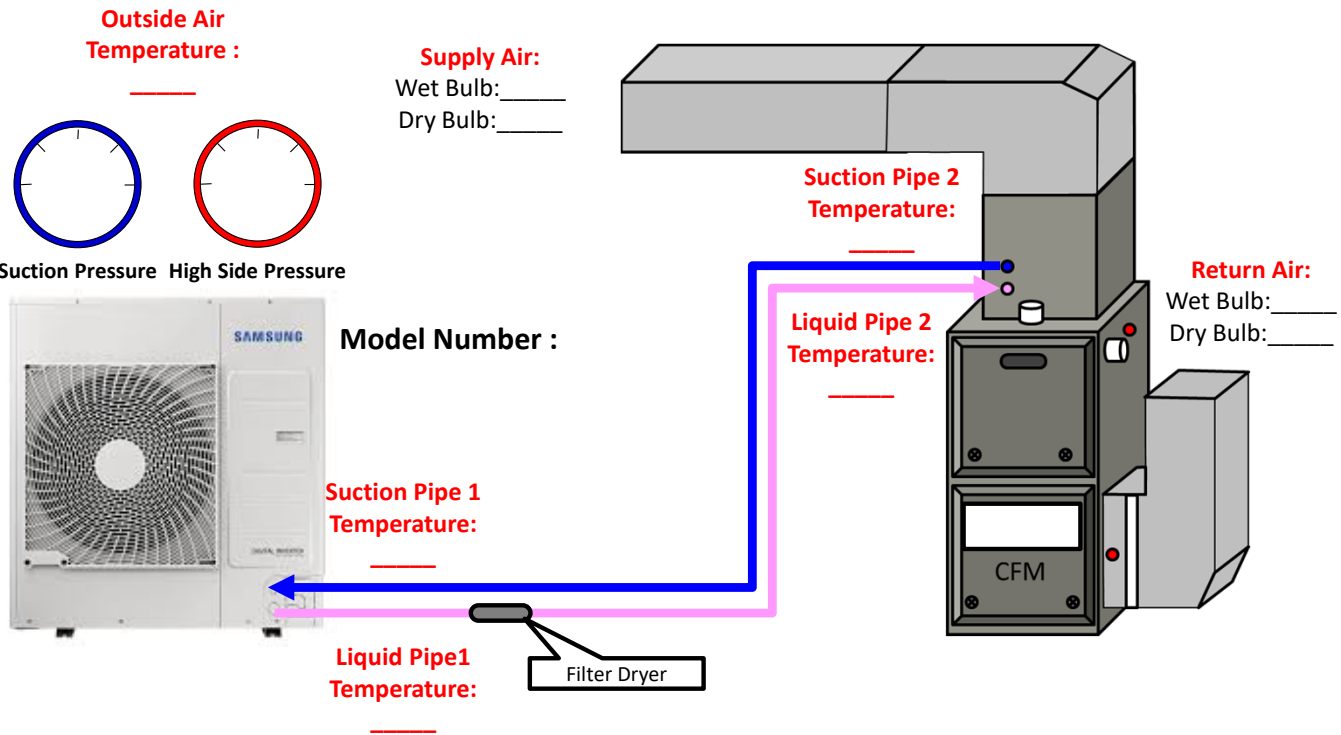
Indoor Delta T: _____
Return Air Temperature _____
Minus
Supply Air Temperature _____

Suction Line:
Size _____
Length _____

Voltage Readings:
ODU
L1 – L2 _____
ACU Module
R to C terminal _____
G to C terminal _____
Y1 to C terminal _____
Y2 to C terminal _____

Liquid Line:
Size _____
Length _____

Directions:
Complete the following form and return to your RTSM or STS. Some information can be obtained using the K 4 Buttons on the outdoor unit and others will need to be taken manually. Air Flow MUST BE VERIFIED for proper operation.



Give a brief description of the system issue:

Installation and Set Up Verification

Hylex Set up Wizzard Questions and Options

<https://www.samsunghvac.com/unitary-settings>



Scan the QR code to go to the hylex set up tool to determine the correct dip switch setting for the installed system. Be prepared to answer the following questions.

What type of indoor unit is installed?

Furnace – Air Handler

Is this a dual fuel application?

Yes - No

Do you want to enable cold air prevention?

Yes - No

Fan operation during defrost?

Yes - No

Configure fan On/Off delay during defrost and cold air prevention?

5 – 10 – 15 seconds

Configure the desired fan step control?

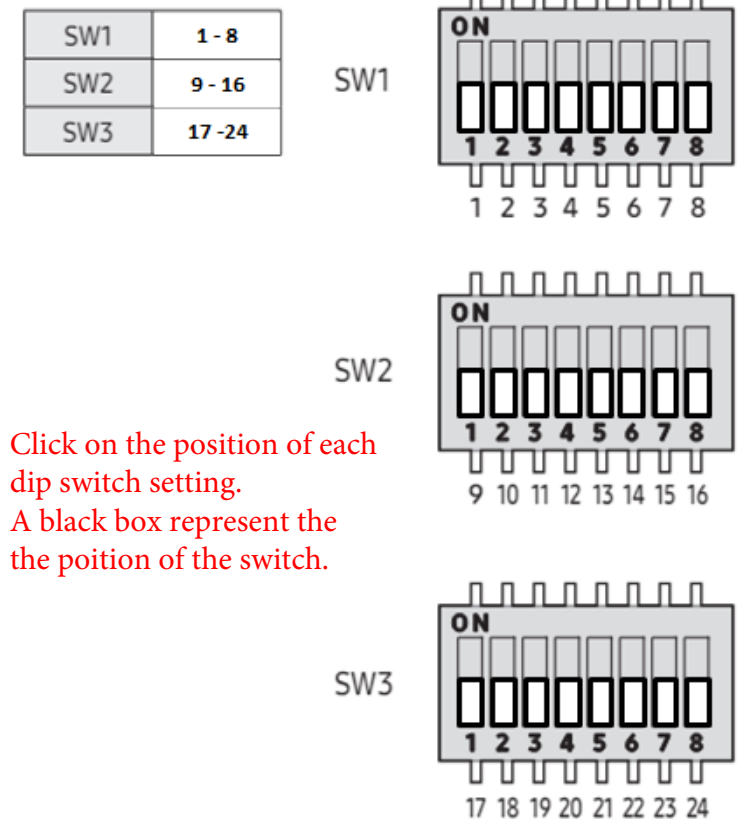
Disabled – 3 step (Y1 only) – 3 Step (Y1 & Y2)

Enable enhanced dehumidification?

Yes - No

Configure the Fan Off delay timer when system turns Off?

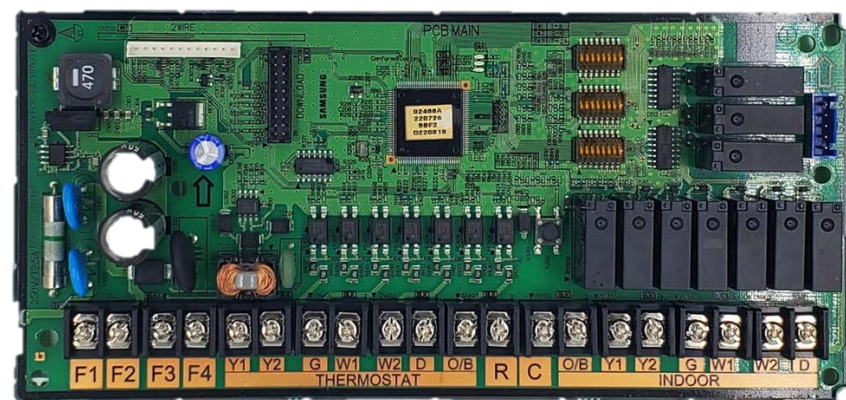
No Delay – 30 – 60 – 90 Seconds



Air Flow Set Up and Verification

The CFM of the system MUST be set and verified.
The measured ESP of the system must not exceed the Max ESP of the motor.

The ESP of the fan is the amount of pressure the fan creates to overcome system resistances.
ESP is the sum of the suction and discharge pressures of the fan and is measured at the inlet of the fan and outlet of the furnace (before the A coil).
Check with the manufacture to determine the correct ESP and testing locations.



Click on each terminal that a wire is connected to.
A circle in the box represent a wire being landed on that terminal

Gathering Service Data

To view operating data from the system you will need to use a combination of the K 4 buttons and typical files service tools like voltmeters, Thermometers, and gauges.

Use the K 4 button to view the following operating data.

It is good practice to check if the system is in protection mode by pushing the K 4 button 13 times.

Because the system operates to hit a target pressure. Pressing the K 4 button 20 times will display the system pressure used for controlling the compressor.

The target pressure will vary depending on the system and call from the thermostat.

Always verify that the ACU has a call for first stage and or 2 stage heating or cooling.

Use this voltage reading and the unit model number to determine the compressor target operations.

Default targets are listed below.

Cooling

AC024BXUPCH/AA		AC036BXUDCH/AA		AC036BXUPCH/AA		AC048BXUPCH/AA		AC060BXUPCH/AA	
Stage2	Stage1	Stage2	Stage1	Stage2	Stage1	Stage2	Stage1	Stage2	Stage1
9.5	10.5	9	10	9	10	8.5	9.5	8.5	9.5
135	150	128	140	128	140	121	135	121	135

Heating

AC024BXUPCH/AA		AC036BXUDCH/AA		AC036BXUPCH/AA		AC048BXUPCH/AA		AC060BXUPCH/AA	
Stage2	Stage1	Stage2	Stage1	Stage2	Stage1	Stage2	Stage1	Stage2	Stage1
26	25	26	25	26	25	28	27	25	24
370	256	370	256	370	256	398	384	256	341

K 4 Service Buttons

K4 short push	Display contents	SEG1	SEG2	SEG3	SEG4	Unit
1	Order frequency	1	Hundreds digit	Tens digit	Units digit	Hz
2	Current frequency	2	Hundreds digit	Tens digit	Units digit	Hz
3	The number of preset indoor units	3	Hundreds digit	Tens digit	Units digit	EA
4	Ambient temperature sensor	4	Hundreds digit or "-" 1)	Tens digit	Units digit	°F or °C 2)
5	Compressor discharge sensor	5	Hundreds digit	Tens digit	Units digit	°F or °C 2)
6	Eva-Mid sensor	6	Hundreds digit or "-" 1)	Tens digit	Units digit	°F or °C 2)
7	Condenser sensor	7	Hundreds digit or "-" 1)	Tens digit	Units digit	°F or °C 2)
8	Current	8	Tens digit	Units digit	The first place of decimals	A
9	Outdoor fan RPM	9	Thousands digit	Hundreds digit	Tens digit	rpm
10	Target discharge temperature	A	Hundreds digit or "-" 1)	Tens digit	Units digit	°F or °C 2)
11	EEV	B	Hundreds digit	Tens digit	Units digit	step
12	The capacity sum of indoor units	C	Tens digit	Unit digit	The first place of decimals	kBtu/h or kW 3)
13	Protective control	D	0: Cooling 1: Heating	Protective control 0: No Protective control 1: Freezing 2: Non-stop defrosting 3: Over-load 4: Discharge 5: Total electric current	Frequency status 0: Normal 1: Hold 2: Down 3: Up_limit 4: Down_limit	-
14	IPM temperature	E	Hundreds digit or "-" 1)	Tens digit	Units digit	°F or °C 2)
15	The number of connected indoor units	F	0	Tens digit	Units digit	EA
16	ESC EEV(CAM)	G	Hundreds digit	Tens digit	Units digit	step
17	ESC IN sensor	H	Hundreds digit or "-" 1)	Tens digit	Units digit	°F or °C 2)
18	ESC OUT sensor	I	Hundreds digit or "-" 1)	Tens digit	Units digit	°F or °C 2)
19	Suction sensor	J	+ / - Hundreds for Fahrenheit	Tens digit	Units digit	°F or °C 2)
20	Pressure sensor	K	+ / - Hundreds for Fahrenheit	Tens digit	Units digit	psi or kg/cm ² 4)

Adjusting Target Values

Before adjusting the target values evaluate the system installation, set up, air flow and charge to verify they are all set correctly.

If the system lacking in performance and all other items above have been verified, you can change the target operation to increase the system performance.

Remember that changing the target pressure will change the saturation temperature of the IDU.

If air flow is not correctly set changing these values will not change system performance.

How to enter the ODU Option Settings:

Press and hold the K 2 button until all 0's appear on the LED Display.

Seg 1 and Seg 2 will display the number for selected option.

Seg 3 and Seg 4 will display the number for setvalue of the selected option.

Press K 1 to change the selected option

Press K 2 to set the desired value of the selected option.

Press and hold K 2 until The LED Display flashes to save the setting.

Option 09 is used to change the target Low pressure in cooling mode.

Option 10 is used to change the target high side pressure in heating mode.

All available options and their available setting can be found in the installation manual located at: <https://www.samsunghvac.com>.

ODU Option 09			Target pressure (psi) based on low pressure correction setpoint									
			AC024BXUPCH/AA		AC036BXUDCH/AA		AC036BXUPCH/AA		AC048BXUPCH/AA		AC060BXUPCH/AA	
			Stage2	Stage1	Stage2	Stage1	Stage2	Stage1	Stage2	Stage1	Stage2	Stage1
Target low pressure calibration for cooling	0(Default)	-	135	149	128	142	128	142	121	135	121	135
	1	- 2.84 psi (- 0.2 kg/cm2)	132	147	125	139	125	139	118	132	118	132
	2	- 5.69 psi (- 0.4 kg/cm2)	129	144	122	137	122	137	115	129	115	129
	3	- 8.53 psi (- 0.6 kg/cm2)	127	141	119	134	119	134	114	127	114	127
	4	- 11.38 psi (- 0.8 kg/cm2)	124	138	117	131	117	131	114	124	114	124
	5	- 14.22 psi (- 1.0 kg/cm2)	121	135	114	128	114	128	114	121	114	121
	6	+ 2.84 psi (+ 0.2 kg/cm2)	138	152	131	145	131	145	124	138	124	138
	7	+ 5.69 psi (+ 0.4 kg/cm2)	141	155	134	148	134	148	127	141	127	141
	8	+ 8.53 psi (+ 0.6 kg/cm2)	144	158	137	151	137	151	129	144	129	144
	9	+ 11.38 psi (+ 0.8 kg/cm2)	147	161	139	154	139	154	132	147	132	147
	10	+ 14.22 psi (+ 1.0 kg/cm2)	149	164	142	156	142	156	135	149	135	149

ODU Option 10			Target High Side Pressure (psig)									
			AC024BXUPCH/AA		AC036BXUDCH/AA		AC036BXUPCH/AA		AC048BXUPCH/AA		AC060BXUPCH/AA	
			Stage2	Stage1	Stage2	Stage1	Stage2	Stage1	Stage2	Stage1	Stage2	Stage1
Target high pressure calibration for heating	0(Default)	-	370.0	256.0	370.0	256.0	370.0	256.0	398.0	256.0	341.0	256.0
	1	+ 2.84 psi (- 0.2 kg/cm2)	372.8	253.8	372.8	253.8	372.8	253.8	400.8	253.8	343.8	253.8
	2	+ 5.69 psi (- 0.4 kg/cm2)	375.7	261.7	375.7	261.7	375.7	261.7	403.7	261.7	346.7	261.7
	3	+ 8.53 psi (- 0.6 kg/cm2)	378.5	264.5	378.5	264.5	378.5	264.5	406.5	264.5	349.5	264.5
	4	+ 11.38 psi (- 0.8 kg/cm2)	381.4	267.4	381.4	267.4	381.4	267.4	409.4	267.4	352.4	267.4
	5	+ 14.22 psi (- 1.0 kg/cm2)	384.2	270.2	384.2	270.2	384.2	270.2	412.2	270.2	355.2	270.2
	6	- 2.84 psi (+ 0.2 kg/cm2)	367.2	253.2	367.2	253.2	367.2	253.2	395.2	253.2	338.2	253.2
	7	- 5.69 psi (+ 0.4 kg/cm2)	364.3	250.3	364.3	250.3	364.3	250.3	392.3	250.3	335.3	250.3
	8	- 8.53 psi (+ 0.6 kg/cm2)	361.5	247.5	361.5	247.5	361.5	247.5	398.5	247.5	332.5	247.5
	9	- 11.38 psi (+ 0.8 kg/cm2)	358.6	244.6	358.6	244.6	358.6	244.6	386.6	244.6	329.6	244.6
	10	- 14.22 psi (+ 1.0 kg/cm2)	355.8	241.8	355.8	241.8	355.8	241.8	383.8	241.8	326.8	241.8